Electromagnetic Wave Optics

Part II: Light Behavior and Properties

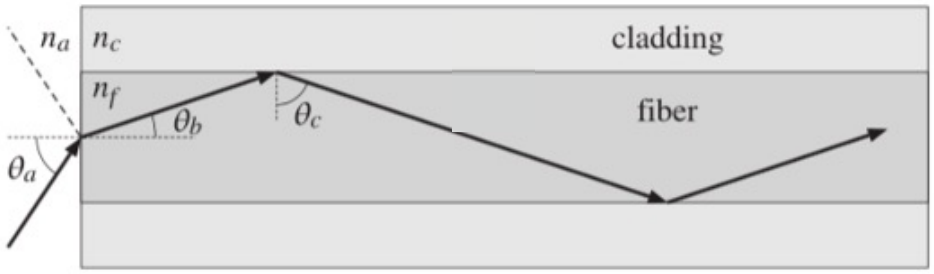
The following subjects will be covered

- Reflection and Transmission of Light.
- Light Polarization.
- Light-matter Interaction.
- Scattering of Light.

Examples

The Optical Fiber.

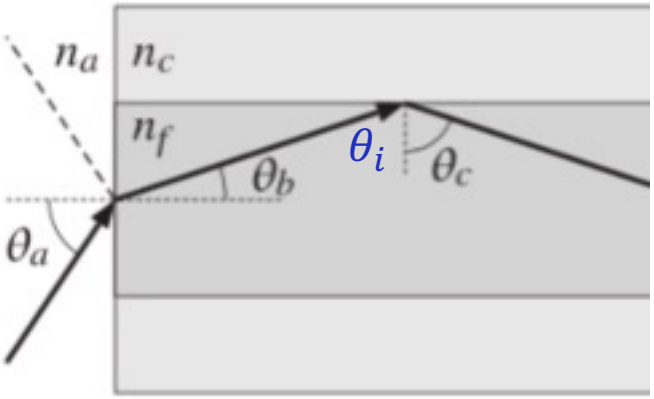
The acceptance angle θ_a should correspond to the critical angle inside the fiber at the cladding-fiber interface. $\theta_i = \theta_c = 90 - \theta_b$



Snell's Law $n_a \sin\theta_a = n_f \sin\theta_b$

$$\sin\theta_a = \frac{n_f}{n_a}\sin\theta_b = \frac{n_f}{n_a}\sin(90-\theta_c)$$

$$\sin\theta_a = \frac{n_f}{n_a}\cos\theta_c = \frac{n_f}{n_a}\sqrt{1-(\sin\theta_c)^2}$$



Critical angle condition

$$\sin\theta_c = \frac{n_c}{n_f} \Rightarrow \sin\theta_a = \frac{\sqrt{(n_f)^2-(n_c)^2}}{n_a}$$

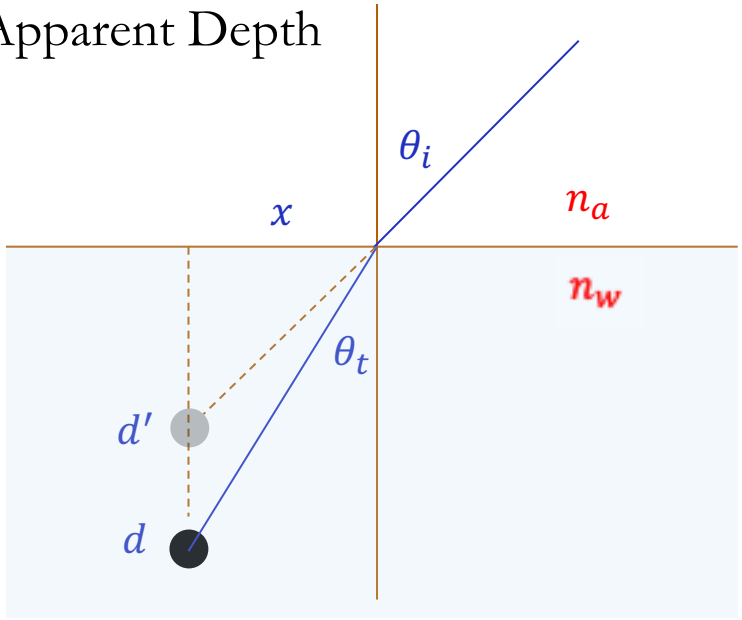
$n_a = 1, n_f = 1.49, n_c = 1.48$

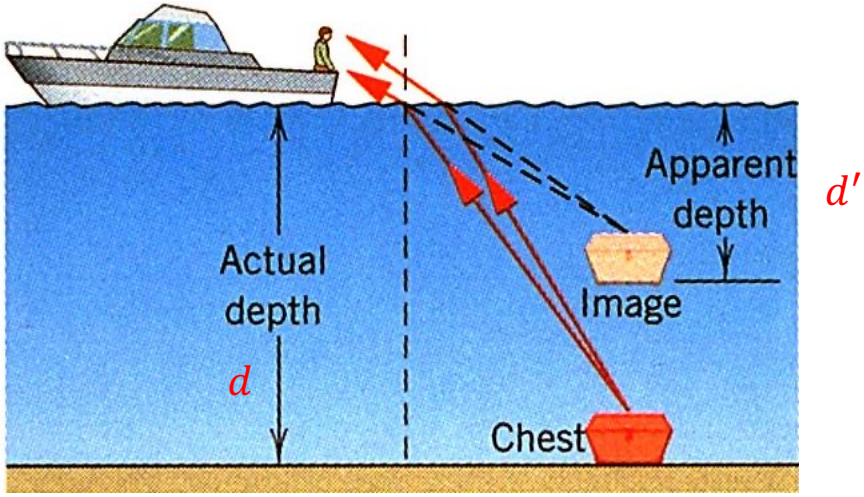
$\theta_c = 83.4°$

$\theta_a = 9.9°$

Examples

Apparent Depth





$$d = x \cot\theta_t$$

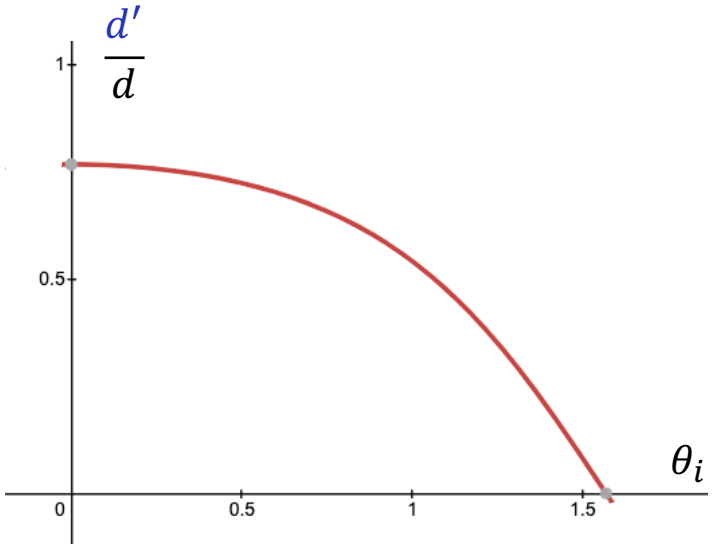
$$d' = x \cot\theta_i$$

$$n_a \sin\theta_i = n_w \sin\theta_t$$

$$\frac{d'}{d} = \frac{\cot\theta_i}{\cot\theta_t} = \frac{\cos\theta_i}{\cos\theta_t} \cdot \frac{\sin\theta_t}{\sin\theta_i}$$

$$\frac{d'}{d} = \frac{n_a}{n_w} \cdot \frac{\cos\theta_i}{\cos\theta_t}$$

$$\frac{d'}{d} = \frac{\cos\theta_i}{\sqrt{n_w^2 - (\sin\theta_i)^2}}$$

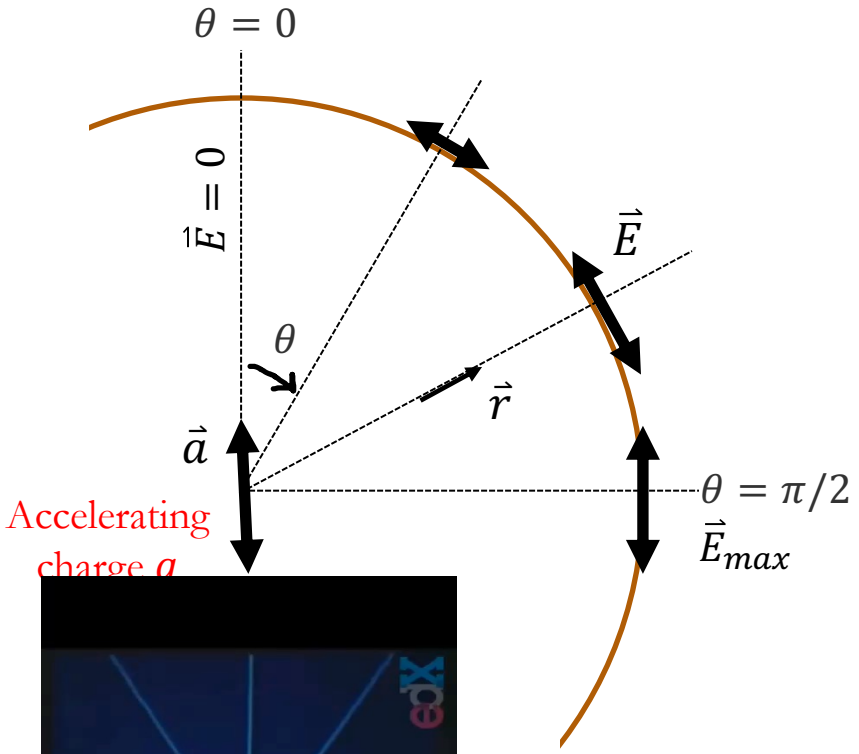


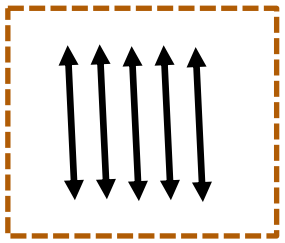
EM Wave Polarization

The polarization of the EM wave is a description of the orientation of the electric field $\vec{E}$. That, in return, is determined by the way the wave (field) is produced.

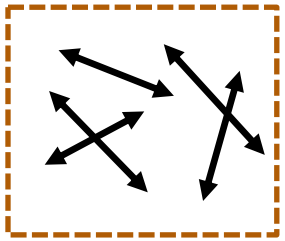
$$\vec{E} \propto q \frac{a}{r} \sin\theta \, \vec{a}_E$$

$$\vec{S} \propto q^2 \frac{a^2}{r^2} . \sin^2\theta \, \vec{a}_r$$





Polarized light



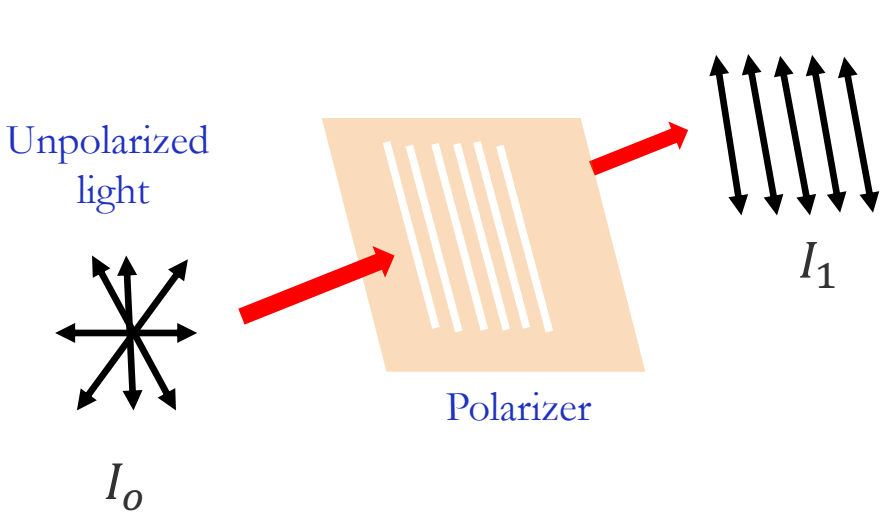
Unpolarized light

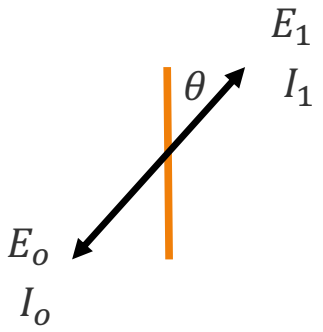


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arized light.

npolarized light?

EM Wave Polarization: Polarizers





$$E_1 = E_o cos\theta$$

$$I_1 = I_o \cos^2\theta$$

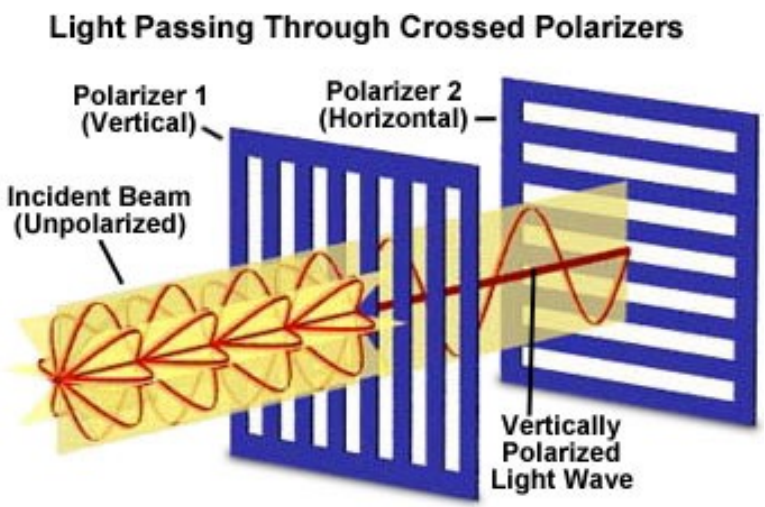
Malus' Law

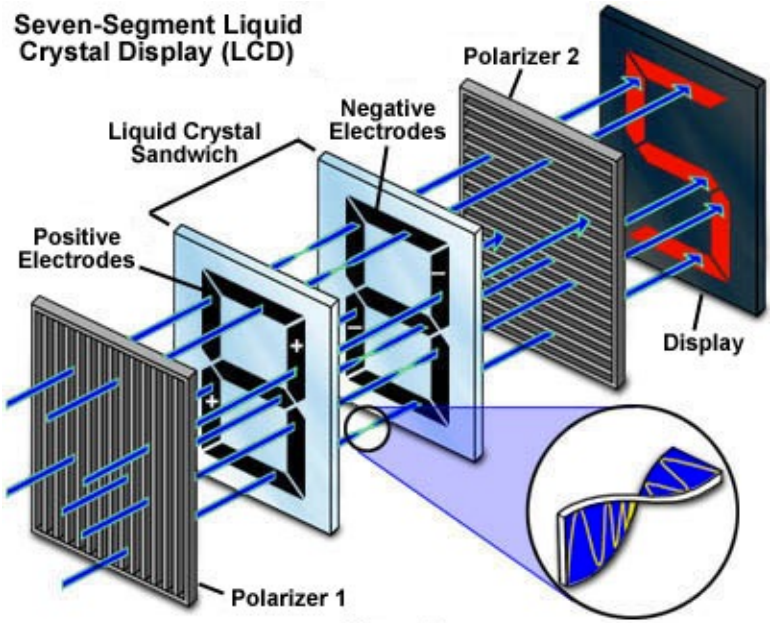
Polarizer output
Integrate over all angles

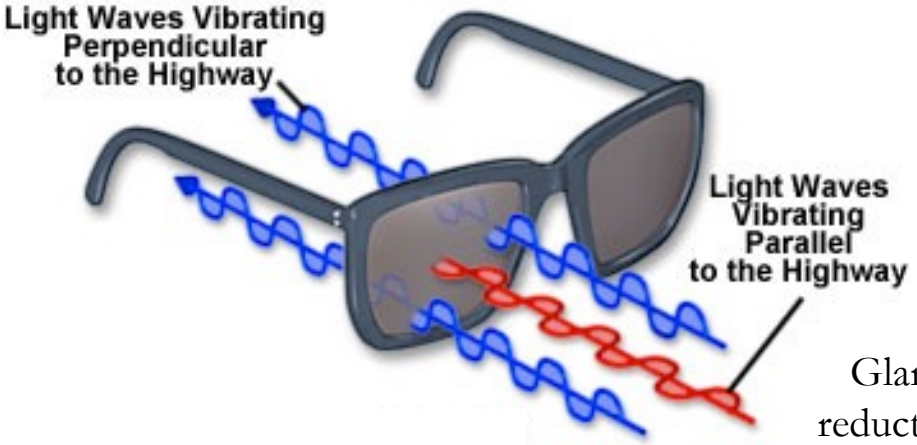
$$I_1 = \frac{1}{2} I_o$$

1) Linear polarization.
2) Intensity goes down.
3) Light color changes??

EM Wave Polarization: Polarizers

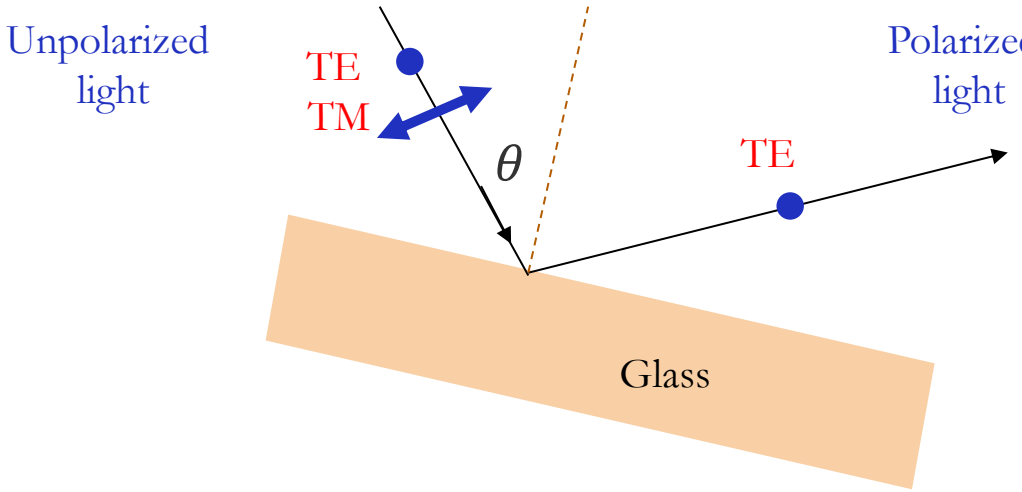






Glare reduction

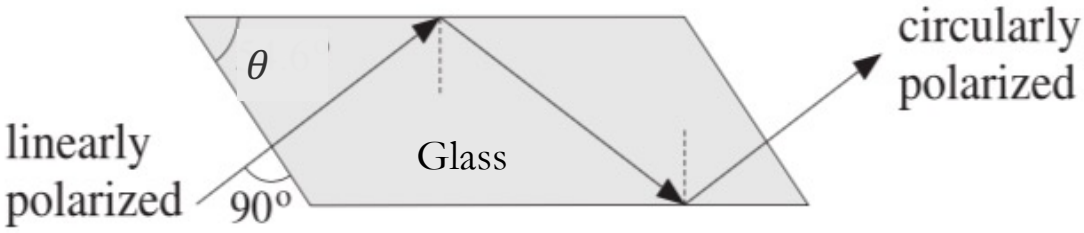
EM Wave Polarization: Polarizers



To polarize light without a polarizer, Find the angle.

Application?

Fresnel Rhomb



Adjust the angle of the rhomb θ such that the output beam is circularly polarized.

For glass with n=1.52, two angles can satisfy the requirement $\theta = 55.46^{o}$ and 47.55^{o}

EM Wave Polarization

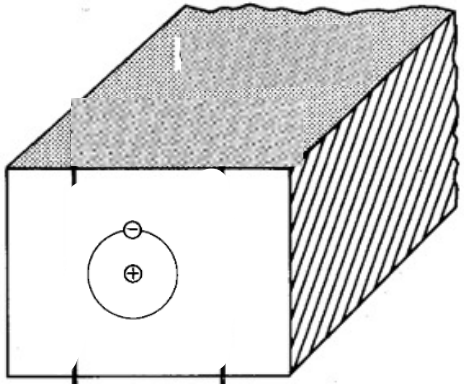
Example 1. Calculate the emerging light intensity when a third polarizer is placed at 45° between two cross polarizers. The incident light is unpolarized with intensity of 600 W/m^2.

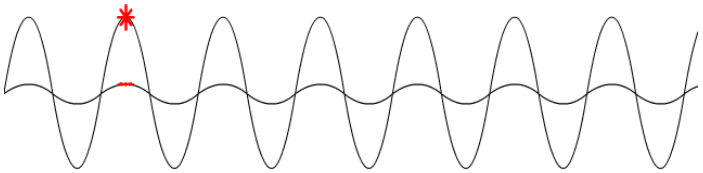
$I_{out} = 75\ W/m^2$

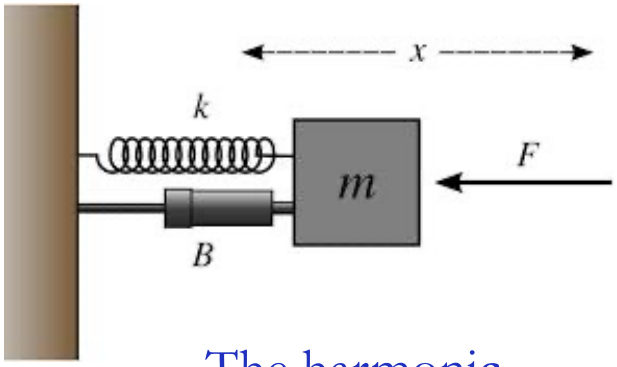
Example 2. Light that emerges out of two polarizers has intensity 65% of the incident light. Is the incident light linearly polarized, circularly polarized or unpolarized?

Example 3. Some materials (such as plastic) when placed between two cross-polarizers, can cause light transmission. Interestingly, these materials interact with different wavelengths differently. For a sheet with non-uniform thickness (possible because of fabrication or stress), what output light do we expect?

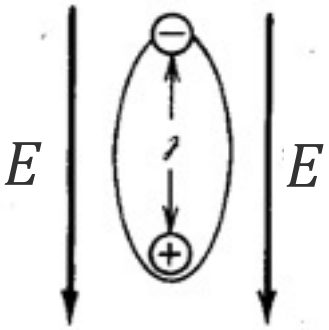
Light-matter Interaction



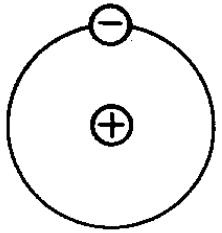




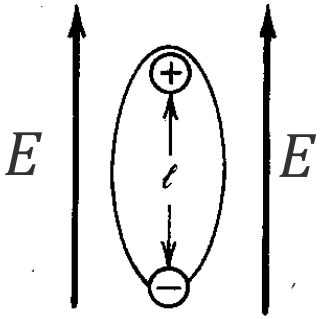
The harmonic oscillator



Atom polarized

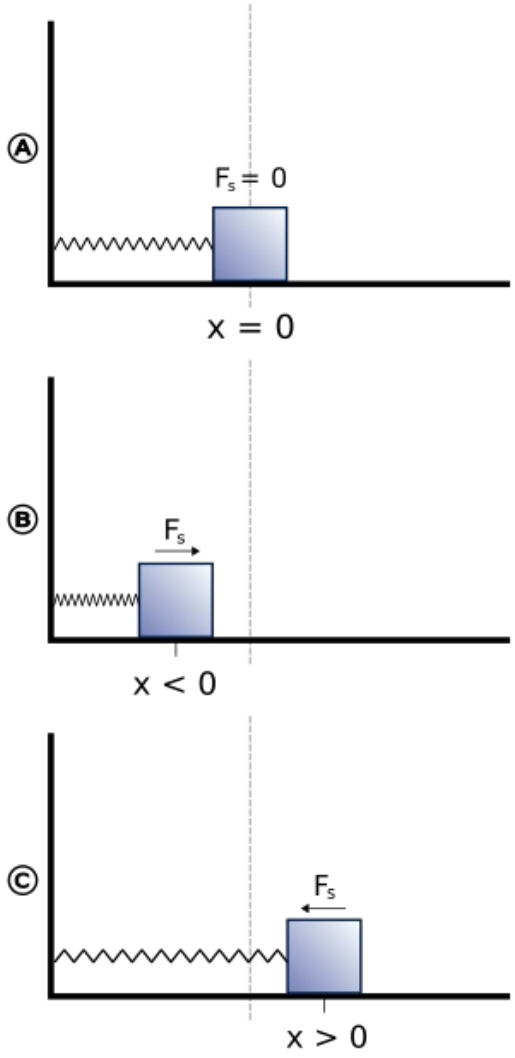


No Wave



Atom polarized

Light-matter Interaction



$$m\frac{\partial^2\vec{x}}{\partial t^2} + m\Gamma\frac{\partial\vec{x}}{\partial t} + m\omega_0^2\vec{x} = -q\vec{E}$$

- $m\frac{\partial^2\vec{x}}{\partial t^2}$: acceleration force
- $m \Rightarrow m_e$: mass of an electron
- $m\Gamma\frac{\partial\vec{x}}{\partial t}$: frictional force
- Γ: damping rate (loss/sec)
- $m\omega_0^2\vec{x}$: restoring force
- $\omega_0 = \sqrt{\frac{K}{m}}$: natural frequency
- $-q\vec{E}$: electric force

Fourier transform

$$m(-j\omega)^2\vec{x}(\omega) + m\Gamma(-j\omega)\vec{x}(\omega) + m\omega_0^2\vec{x}(\omega) = -q\vec{E}(\omega)$$

Simplify

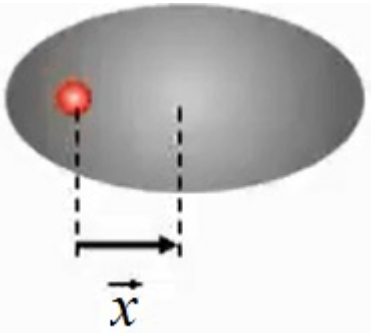
$$\left(-m\omega^2 - j\omega m\Gamma + m\omega_0^2\right)\vec{x}(\omega) = -q\vec{E}(\omega)$$

Light-matter Interaction

$$\vec{x}(\omega) = -\frac{q}{m_e}\frac{\vec{E}(\omega)}{\omega_0^2 - \omega^2 + j\omega\Gamma}$$

$$\vec{P}(\omega) = N\left\langle \vec{p}(\omega) \right\rangle = \varepsilon_0 \chi(\omega)\vec{E}(\omega)$$

$$\chi(\omega) = \left(\frac{Nq^2}{\varepsilon_0 m_e}\right)\frac{1}{\omega_0^2 - \omega^2 + j\omega\Gamma}$$



$$\vec{p}(\omega) = -q\vec{x}(\omega)$$

$$\chi(\omega) = \frac{\omega_p^2}{\omega_0^2 - \omega^2 + j\omega\Gamma}$$

Lorentz Model

$$\omega_p^2 = \frac{Nq^2}{\varepsilon_0 m_e}$$ plasma frequency

$q = 1.60217646\times10^{-19}$ C

$\varepsilon_0 = 8.8541878176\times10^{-12}$ F/m

$m_e = 9.10938188\times10^{-31}$ kg

Light-matter Interaction

$$\varepsilon_r(\omega) = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - j\omega\Gamma} \qquad \omega_p^2 = \frac{Nq^2}{\varepsilon_0 m_e}$$

$$\varepsilon_r(\omega) = \varepsilon_r'(\omega) - j\varepsilon_r''(\omega)$$

$$\varepsilon_r'(\omega) = 1 + \omega_p^2 \frac{\omega_0^2 - \omega^2}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\Gamma^2}$$

$$\varepsilon_r''(\omega) = \omega_p^2 \frac{\omega\Gamma}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\Gamma^2}$$

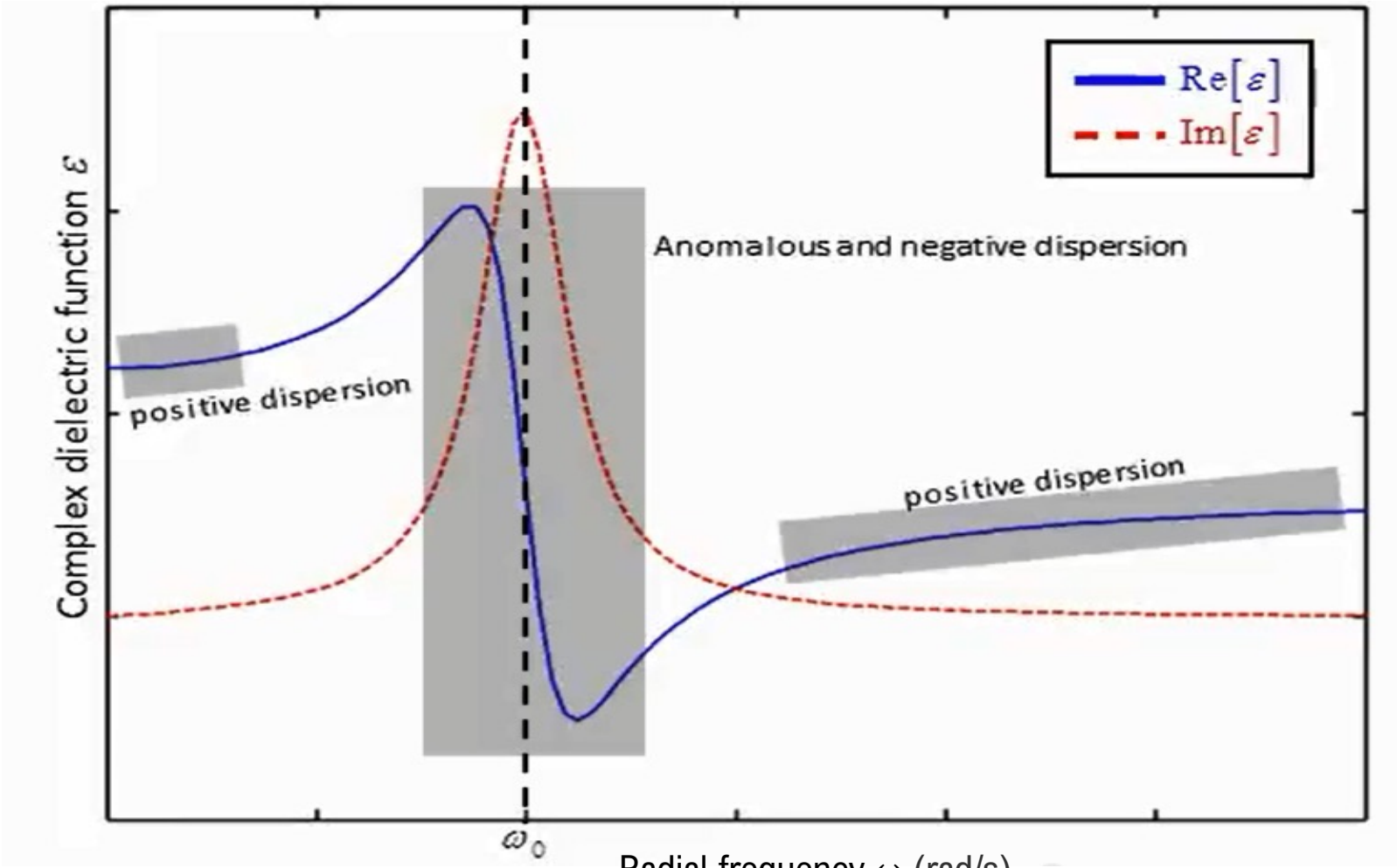
$$n = \sqrt{\mu_r \varepsilon_r} = n' - jn''$$ Complex Refractive index

$$(n')^2 = \frac{\varepsilon_r'}{2} + \frac{1}{2}\sqrt{(\varepsilon_r')^2 + (\varepsilon_r'')^2}$$

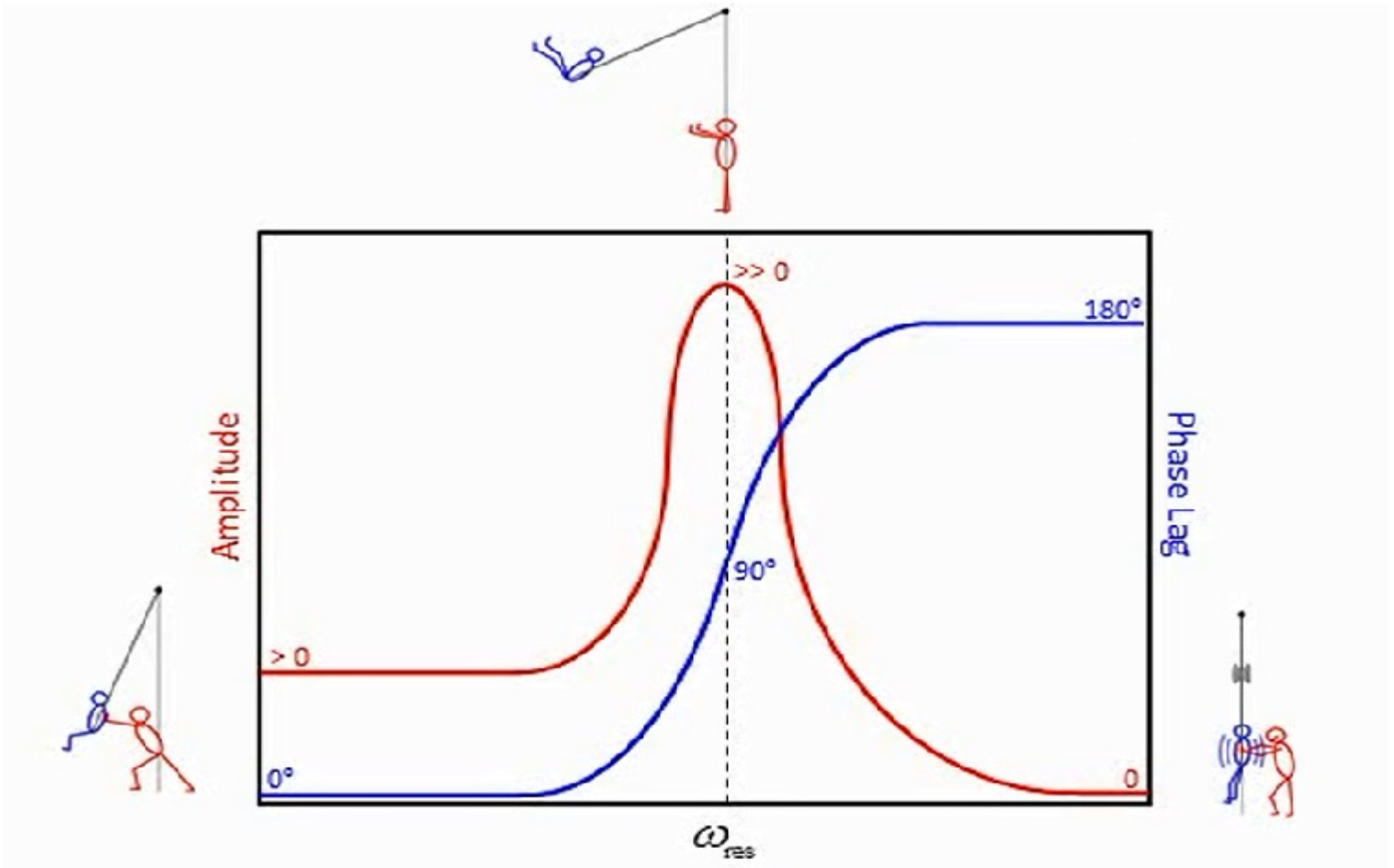
$$n'' = -\frac{\varepsilon_r''}{2n'}$$ Extinction coefficient

$$\alpha = 2\frac{\omega}{c}n''$$ Absorption Coefficient

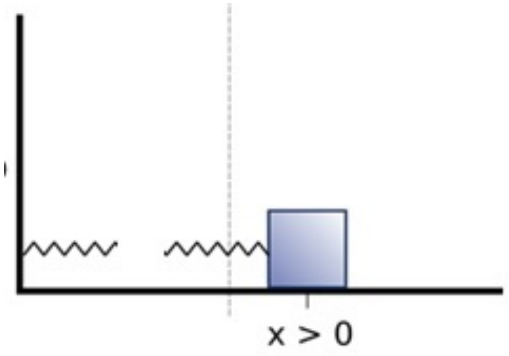
Light-matter Interaction: Lorentz Model



Light-matter Interaction: Resonance



Light-matter Interaction: Metals



$$m\frac{\partial^2\vec{x}}{\partial t^2}+m\Gamma\frac{\partial\vec{x}}{\partial t}+m\omega_0^2\vec{x}=-q\vec{E}$$

acceleration force

$m \Rightarrow m_e$ mass of an electron

frictional force

Γ damping rate (loss/sec)

restoring force

$\omega_0=\sqrt{\frac{K}{m}}$ natural frequency

electric force

$$\varepsilon_r(\omega)=1-\frac{\omega_p^2}{\omega^2-j\omega\Gamma}$$

$$\varepsilon_r=\left(1-\frac{\omega_p^2\tau^2}{1+\omega^2\tau^2}\right)+j\left(\frac{\omega_p^2\tau/\omega}{1+\omega^2\tau^2}\right)$$

Drude Model

$$n=\sqrt{\mu_r\varepsilon_r}=n'-jn''$$

Complex Refractive index

$$(n')^2=\frac{\varepsilon_r'}{2}+\frac{1}{2}\sqrt{(\varepsilon_r')^2+(\varepsilon_r'')^2}$$

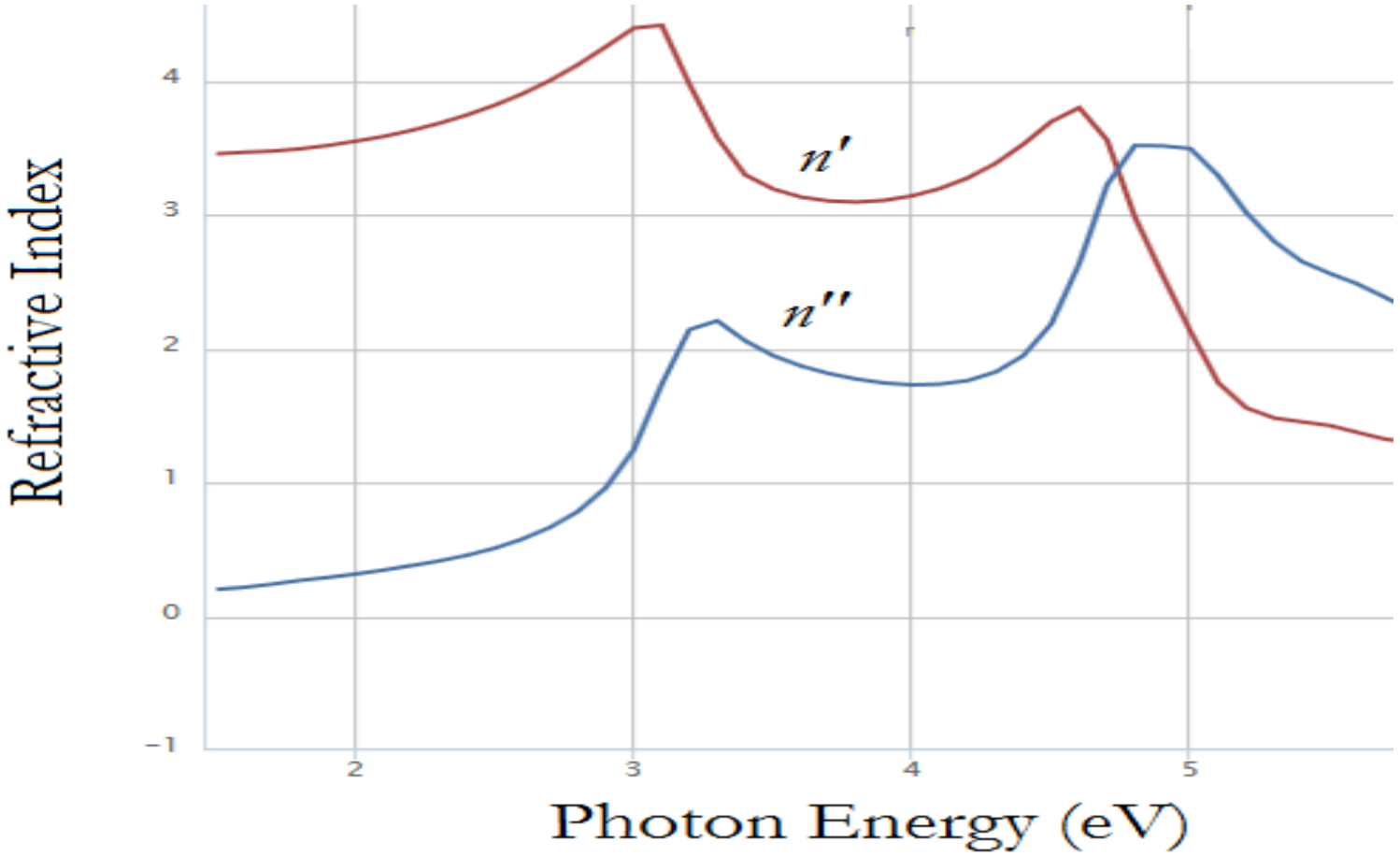
$$n''=-\frac{\varepsilon_r''}{2n'}$$

Extinction coefficient

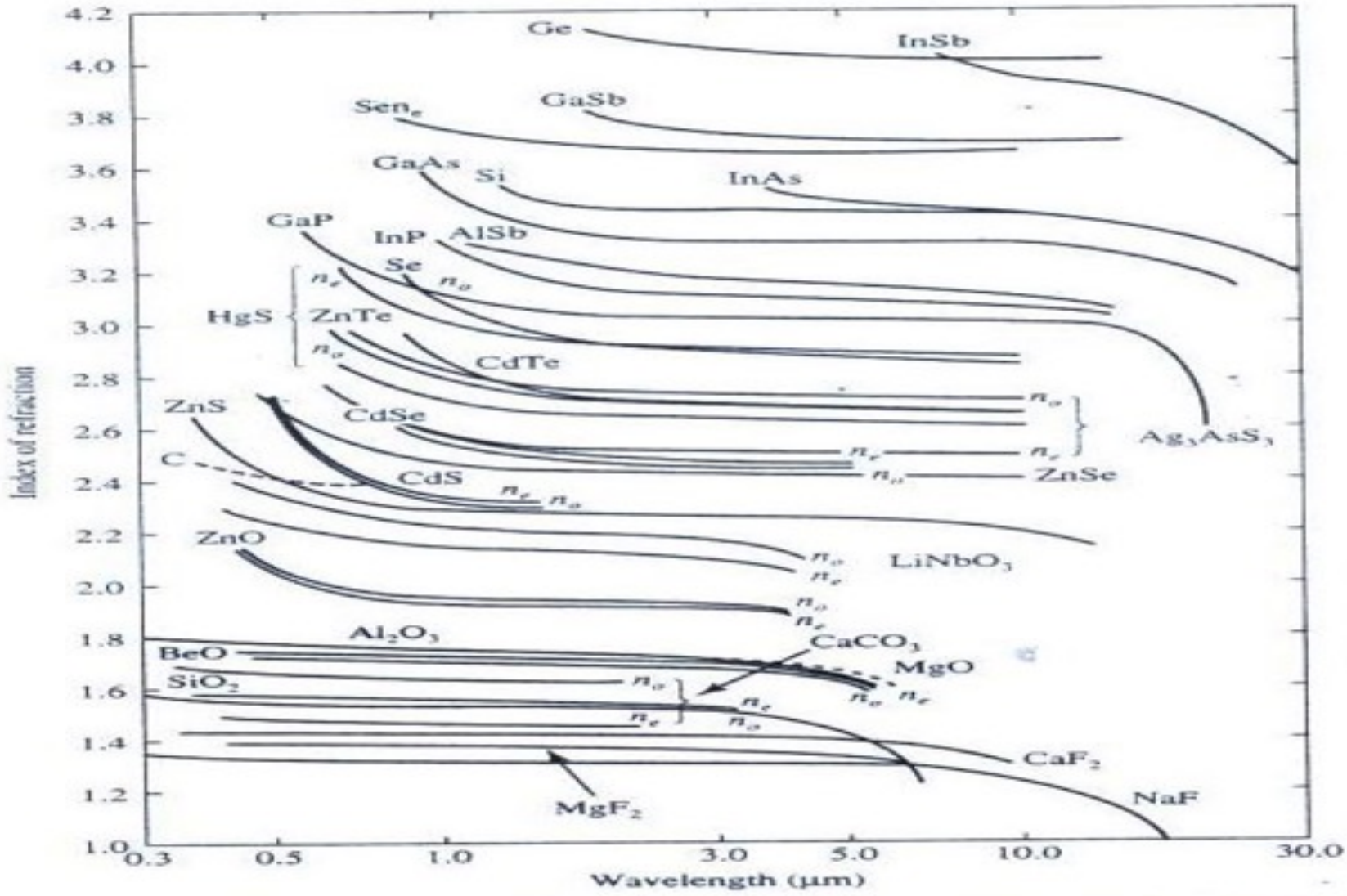
$$\alpha=2\frac{\omega}{c}n''$$

Absorption Coefficient

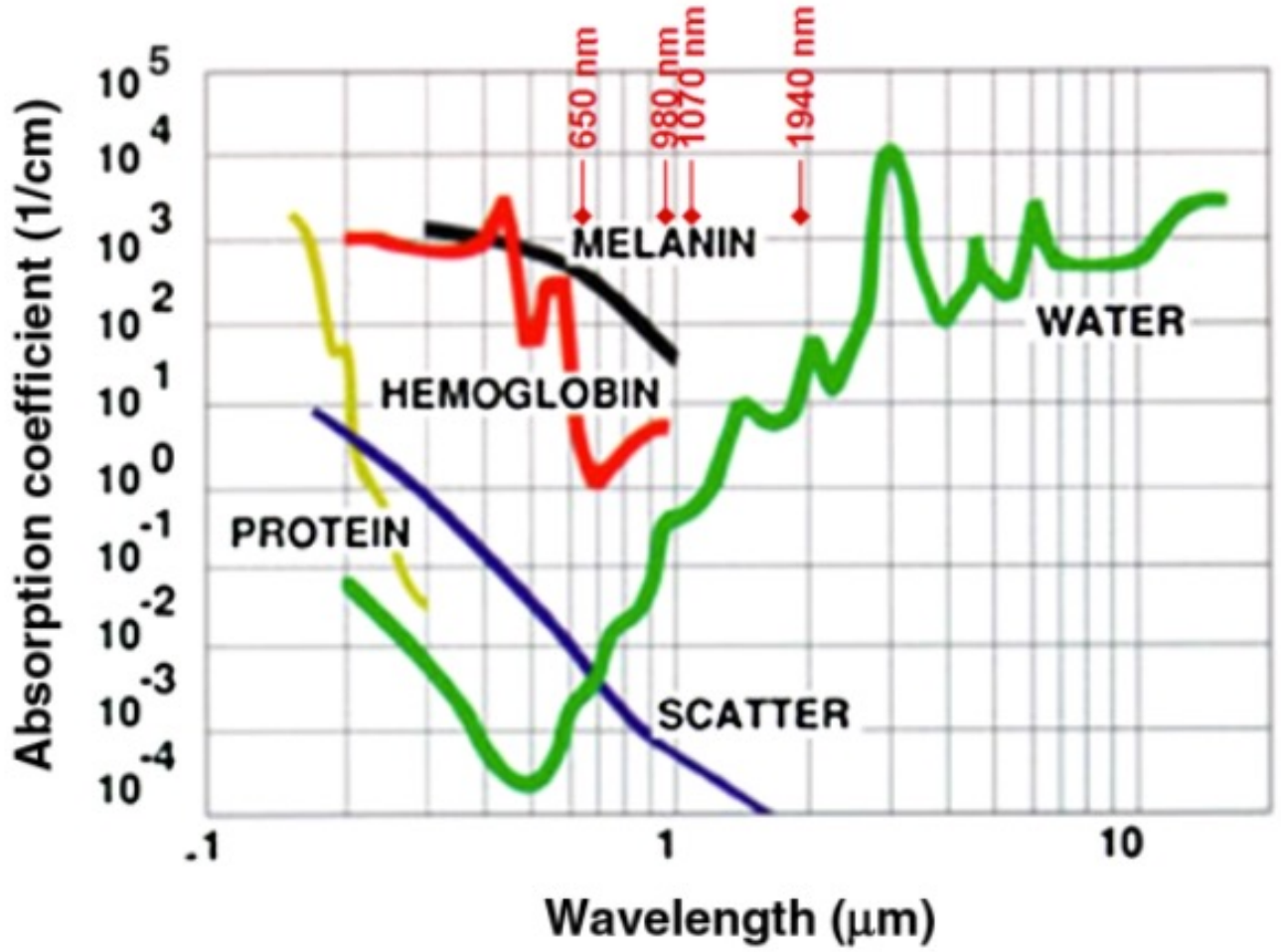
Light-matter Interaction: Examples
Refractive Index of InP



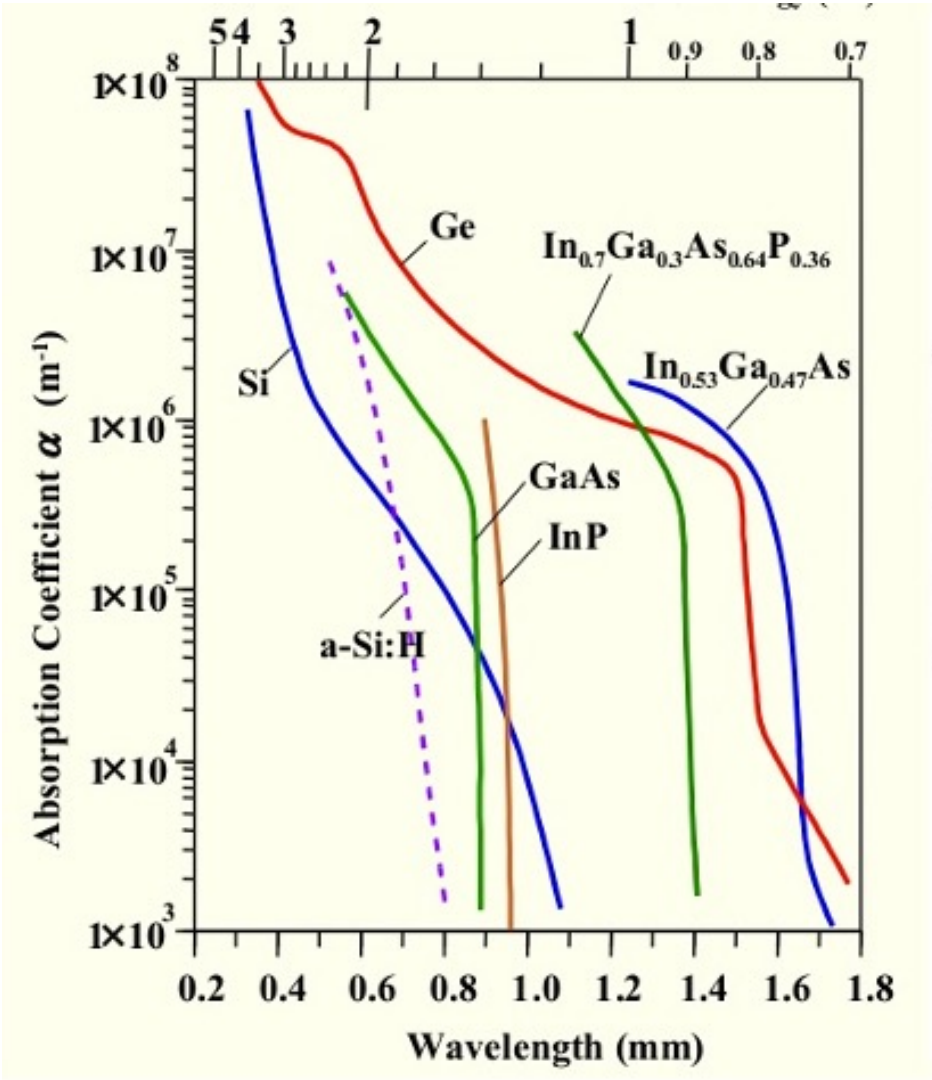
Light-matter Interaction: Examples
Other Semiconductors

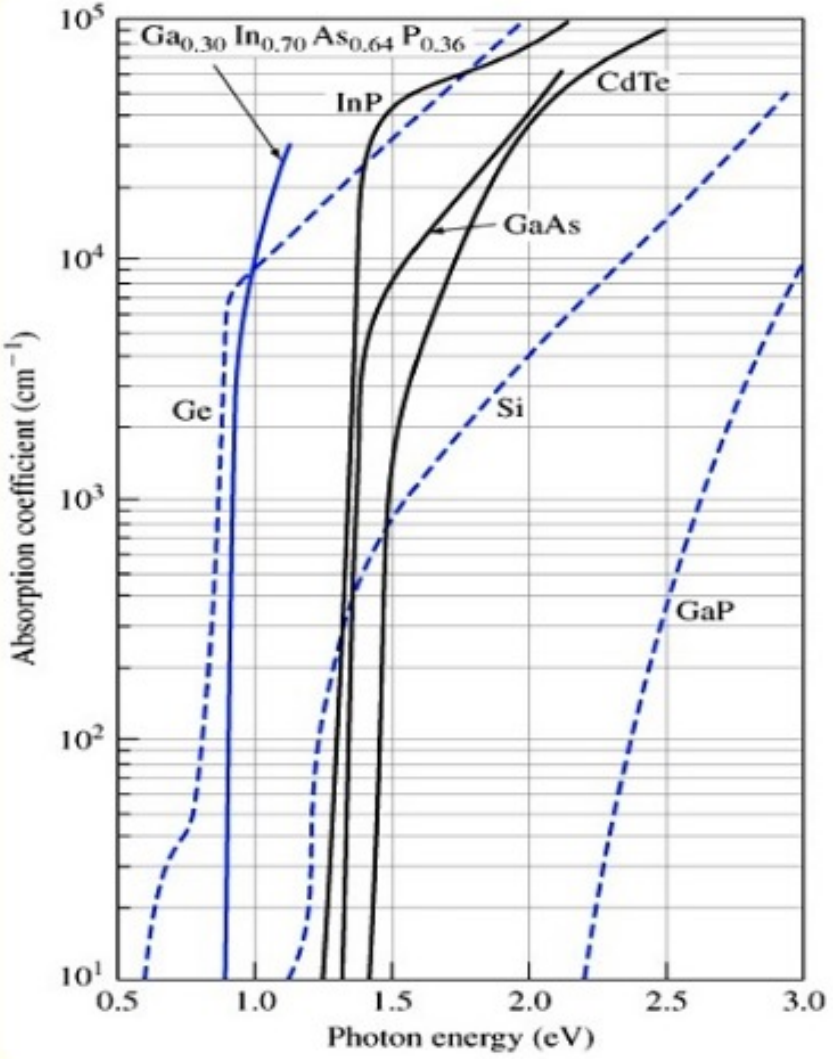


Light-matter Interaction: Examples
Absorption Coefficient



Light-matter Interaction: Examples
Absorption Coefficient





Q & A

Send me an email if you have any questions.